6. Domain: Internet of Things (IoT)

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	Traffic Signal Remote Monitoring & Management Software
	Name of the Project under which technology is developed	InTranSe Phase-I (ITS R&D program)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 9
	Brief Write-up on the Technology Developed	TraMM-EnV is a core component of the Advanced Traffic Management System (ATMS), designed to centrally monitor and manage a network of C-DAC Adaptive Traffic Signal Controllers deployed across city or zonal junctions from an ATMS Command and Control Centre. The platform provides a user-friendly HMI and dashboards for remote configuration, visualization, analysis of traffic patterns, and signal control, with data presented through trends, charts, reports, and live animation screens. TraMM-EnV includes a dedicated user management system with role-based access and alarms, supports centralized junction configuration with remote plan upload/download, and periodically collects real-time junction data from controllers for distribution to associated modules, including the CoSiCoSt Adaptive Traffic Control System, for display, reporting, and processing.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Benoy Gopal E. B. , C-DAC Thiruvananthapuram 9447057046, benoy@cdac.in Satheesh G, C-DAC Thiruvananthapuram 9496750350, satheeshg@cdac.in
2.	**Brief details of Technology Developed**	CoSiCoSt is an Adaptive Traffic Control System (ATCS) software algorithm for optimizing signal timings for the typical Indian driving and traffic conditions such as poor lane discipline and high heterogeneity
	Name of the Project under which technology is developed	InTranSe Phase-I (ITS R&D program)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 9
	Brief Write-up on the Technology Developed	CoSiCoSt-EnV is an Adaptive Traffic Control System (ATCS) software algorithm for optimizing signal timings for the typical Indian driving and traffic conditions such as poor lane discipline and high heterogeneity. CoSiCoSt bears an Indian Patent (No: 239258 Title: A Method for Synchronizing Heterogeneous Road Traffic and System thereof) jointly owned by the Ministry of Electronics and Information Technology (MeitY) and C-DAC.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Benoy Gopal E B , 9447057046, benoy@cdac.in Satheesh G, 9496750350, satheeshg@cdac.in C-DAC Thiruvananthapuram

Sr. No.	Parameters	Brief details
3.	**Brief details of Technology Developed**	ATCS compatible Vehicle Actuated Traffic signal controller - CUTE- NG
	Name of the Project under which technology is developed	Transfer of Technology Support (ToT Support)
	Funding Agency	Internal core funding of C-DAC
	Technology Readiness Level (TRL)	9
	Brief Write-up on the Technology Developed	The C-DAC has developed a next-generation hardware platform, CUTE-NG, which is a 64-bit standalone, demand-actuated road traffic signal controller with Area Traffic Control System (ATCS) compatibility. The platform is capable of functioning as either a pre-timed controller or a vehicle actuated controller at isolated intersections. Additionally, it can be integrated into a coordinated signal control system, operating in either pre-timed or vehicle actuated modes. It also supports local level control of an Adaptive Traffic Control System (ATCS), with the ability to control signal lamps over a wired medium.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Prakash R, prakashr@cdac.in Hemant Jeevan Magadum, hemant@cdac.in Satheesh G,9496750350, satheeshg@cdac.in C-DAC Thiruvananthapuram
4.	**Brief details of Technology Developed**	Centralized traffic emergency access routing With Signal Priority (ClearWay)
	Name of the Project under which technology is developed	Emergency Service Vehicle Priority at Signalised Intersections
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	9
	Brief Write-up on the Technology Developed	ClearWay is an emergency vehicle priority management solution, developed as an offshoot of the Vehicle Tracking and Fleet Management Software (*FlexiFleet*). It ensures faster clearance at traffic junctions by leveraging GPS data, geofencing, and integration with traffic signal systems to create green corridors, reducing response times and helping save lives during the critical golden hour. With its cost-effective architecture, ClearWay avoids heavy infrastructure investments by using mobile applications and existing devices for vehicle onboarding, while offering a scalable and flexible deployment model suitable for both small towns and large metropolitan cities. The system comprises a web application, driver mobile app, and integrated modules for real-time tracking, priority zone configuration, route management, alert settings, zone detection, signal interfacing, and reporting. These features collectively provide centralized monitoring, dynamic signal prioritization, and advanced analytics for continuous improvement in emergency response. ClearWay also supports GPS data from AIS-140 VLT devices and offers integration with other fleet

Sr. No.	Parameters	Brief details
		management systems, ensuring seamless adoption without major modifications.
	Details of Contact Person alongwith Mobile No. and e-mail ID	George Thomas, 9895096434, georget@cdac.in Satheesh G, 9496750350, satheeshg@cdac.in C-DAC Thiruvananthapuram
5.	**Brief details of Technology Developed**	C-DAC Kolkata has a product on Smartcard based solution for authentication, attendance, access control and monitoring complying to the Indian Smart Card Operating System standard named SCOSTA. Catering to all the components of a Smart Card based system comprising of Smart Card Readers, Application File structures in cards and an Interactive Application has been designed. We intend to realize the usage of Smart Card based system across diverse needs through flexible configurations supported through an interpreter system. The solution offers the dynamics of various Organizational policies and their geographically distributed offices with heterogeneous infrastructure and resource availability. The basic design philosophy adopted as maintaining the essential functionality and operation of the system in the disrupting environment like Network outage, sudden power failure and communication failure etc. Besides this, The Smartcards are interoperable throughout the Institutional network i.e. if an employee moves from one office or even departments, his/her attendance or access control will be taken care of irrespective of the location. We have come up with a secure design of the smartcard application and user data with using 3DES authentication along with operational restriction up-to the validity period of the card. The security design is one of the unique features and based on the hierarchical model of roles and responsibilities with shared secret to minimize the risk of security compromise.
	Name of the Project under which technology is developed	Centre of Excellence in Smart Card Technologies
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-9
	Brief Write-up on the Technology Developed	The referred indigenous Smart Card & Biometry based identity and access management system solution has the following Novel features: o Based on Open standard (SCOSTA) developed jointly by CDAC and IIT Roorkee o Can support multiple application besides attendance like canteen, access control, desktop and server login o More secured by advanced key management and User PIN and Biometric Authentication o Support both offline / online mode of Operation and Data Storage o UID Verified Finger print Enrollment for the First-time use, also supports exceptions o Contains more information 64K including Photo o Same card can have multiple roles / Authority

Sr. No.	Parameters	Brief details
		o Cryptographic Key Based System Design as per user Security Requirement Real-time information for all User's
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Biswajit Saha, C-DAC Kolkata 9432215536, biswajit.saha@cdac.in
6.	**Brief details of Technology Developed**	Aqua-SURAKSHA, the device is designed to detect organophosphate pesticides (OPs) residues, which are potent Endocrine Disrupting Chemicals (EDCs) harmful to aquatic ecosystems. This system combines an enzyme-based biochemical detection method for EDCs with an image analysis-based intelligent computing algorithm for decision-making. Furthermore, the 'Aqua SURAKSHA' system has undergone laboratory trials, demonstrating strong consistency with results from conventional testing conducted in ten selected water bodies across India. It qualitatively predicts the presence of toxic EDCs, primarily OPs, in water and assesses the safety index, determining whether the water is safe or unsafe for use.
	Name of the Project under which technology is developed	MEAN: Measuring Endocrine Disrupting Chemicals (EDC) and Aquatic diagnostics through Bio-Sensory Network with a special reference to North East India
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL - 9
	Brief Write-up on the Technology Developed	The developed technology is a portable, enzyme-based colorimetric bio-sensing system engineered for the rapid, in-situ detection of Endocrine Disrupting Chemicals (EDCs), specifically Organophosphates (OPs), in water ecosystems. This standalone device functions on a Single Board Computing module (Windows 10) and integrates a specialized Inbuilt Uniform Illumination Imaging System (UIIS) to perform advanced color analysis. It achieves real-time qualitative and quantitative estimation of contaminants with a sample-to-result time of less than 15 minutes. Featuring an inbuilt battery and a Graphical User Interface (GUI), the system operates independently of external power, optimizing it for field deployment. Furthermore, it supports remote connectivity via a mobile dashboard for real-time data monitoring. By synergizing biochemical enzymatic reactions with computational imaging, this technology offers a robust, high-speed solution for environmental surveillance without the need for complex laboratory infrastructure.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr Subhankar Mukherjee, C-DAC Kolkata 9477355692, subhankar.mukherjee@cdac.in
7.	**Brief details of Technology Developed**	**IndusCopter Development Kit:** The indigenously developed micro-UAV/drone development kit can be used to understand drone technology and to develop drone-related applications.
	Name of the Project under which technology is developed	PUSHPAK/Internal

Sr. No.	Parameters	Brief details
	Funding Agency	MeitY/Internal
	Technology Readiness Level (TRL)	8
	Brief Write-up on the Technology Developed	The **IndusCopter Development Kit** is an indigenous micro-UAV/drone development platform powered by C-DAC's IndusCopter flight controller. Its powerful microcontroller and rich set of peripherals make it a suitable platform for developing a wide range of UAV-based applications. The kit also includes an interactive desktop-based configurator tool for easy configuration and data visualization. **Key Features** • Indigenous Micro-UAV Drone • Based on C-DAC's IndusCopter Flight Controller • Integrated Firmware for Telemetry, Flight control and Application • Navigation Control through android app/remote control • Generic UAV platform for various application **Key Use case** • Education and Research • Smart Agriculture • Environmental Monitoring • Surveillance Crowd Monitoring
	Details of Contact Person along with Mobile No. and e-mail ID	Name: Harish Chandra, C-DAC Bangalore Mobile: 8299071407 Email: harishchandra@cdac.in
8.	**Brief details of Technology Developed**	The **UAV Motor Test Kit** is an indigenously developed solution designed to measure and analyse BLDC motor parameters for drone development.
	Name of the Project under which technology is developed	Project: PUSHPAK – National Mission on Drone Technology Towards Drone Excellence. Sub Project: Multi-Functional Drone with In-Built Artificial Intelligence
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	8
	Brief Write-up on the Technology Developed	A **UAV Motor Test Kit** is a device used to measure key BLDC motor parameters, including RPM, thrust, vibration, and current consumption. It is essential for evaluating motor performance, enabling accurate estimation of a drone's total payload capacity and expected flight time. The kit includes an interactive desktop application that displays data graphically and supports detailed analysis of payload capacity and flight duration. **Key Features** • Measures thrust, RPM, vibration and current of BLDC motor • Measured data can be seen in graphical manner in GUI application.

Sr. No.	Parameters	Brief details
		Key Use case • Analyse motor performance and Efficiency • Payload capacity of drone system can be analysed Total runtime of drone can be determined
	Details of Contact Person alongwith Mobile No. and e-mail ID	Name: Harish Chandra, C-DAC Bangalore Mobile: 8299071407 Email: harishchandra@cdac.in
9.	**Brief details of Technology Developed**	AQ-AIMS, a Sensor Based Air Quality Monitoring Device with the following features: • Pre-calibrated gas sensor modules for different gases including O3, SO2, CO, NO2. • Monitors particle concentration of ambient air in PM1, PM2.5, PM5, PM10 • Integrated web server for visualization, configuration, data logging. • Compact size for deployment, installation, and operation in minimal time • Enabled with GSM communication for seamless data transmission • Water Resistant Enclosure for outdoor deployment
	Name of the Project under which technology is developed	National Programme on Electronics and ICT Applications in Agriculture and Environment (AgriEnIcs)
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-8
	Brief Write-up on the Technology Developed	The solution provides a method for monitoring and predicting outdoor live Air Quality Index (AQI) using an AI-based decision-making algorithm and IoT sensors. Air Quality Monitoring System (AQ-AIMS) is a 24x7 real time monitoring system for the outdoor environment which monitors the environmental pollutants like CO, SO2, NO2, O3, CO2 & PM2.5, PM10, PM1 along with temperature & humidity. The system generates Air Quality Index (AQI) as per CPCB guidelines considering the pollutants. The data transmission is being executed by GSM communication to a web server for continuous monitoring, analysis and visualization. The standalone system can be powered through a secondary source of power (Renewable energy i.e solar powered) for sustainable operation. The device can be deployed in Cement and Mines Industries (preferably) or any outdoor environment. A Mechanical enclosure has been also made to sustain the system under a mining environment also. The device may also be deployed across the transportation routes of the mine where live monitoring and predicting capability of Future Air Quality Index plays a pivotal role to control the Air Pollution within a certain range.

Sr. No.	Parameters	Brief details
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr Hena Ray, 9830764590, hena.roy@cdac.in Mr. Sangit Saha, 8667726087/7639811569 (Whatsapp), sangitsaha@cdac.in C-DAC Kolkata
10.	**Brief details of Technology Developed**	A high-throughput, field-portable miniaturized device integrated with stabilized bioreceptors is designed for cost-effective diarrhea prognosis. This device enables the prediction of bacterial causes of diarrhea, thereby reducing unnecessary antibiotic use. Additionally, it features point-of-use capabilities. CoLOSENS has the potential to enhance rural healthcare, particularly in areas where expensive instrumentation is either scarce or unaffordable.
	Name of the Project under which technology is developed	An Affordable Colorimetric Diagnostic Instrument: ColOSENS and Field Validation at Imphal, North-East India
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL - 8
	Brief Write-up on the Technology Developed	This technology constitutes a portable, colorimetric biosensing system engineered for the rapid, onsite diagnosis of the microbial origin of diarrhoea. Integrating a single-board computing module with an image acquisition unit and LED-optimized lighting, the device analyzes samples utilizing stabilized bioreceptors and a specific reagent kit. It delivers precise predictions of bacterial diarrhoea within just 15 minutes, operating independently of external power sources. Designed for Point-of-Care (POC) utility, the system features a user-friendly Android Graphical User Interface (GUI) and a mobile dashboard for real-time data monitoring. This robust architecture renders the device highly suitable for deployment in resource-limited rural health centers and personal home setups, facilitating efficient and accessible disease management.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr Subhankar Mukherjee, C-DAC Kolkata 9477355692, subhankar.mukherjee@cdac.in
11.	**Brief details of Technology Developed**	Air quality monitoring system that continuously measures environmental pollutants such as particulate matter (PM2.5, PM10), carbon dioxide (CO2) along with other atmospheric parameters like light intensity, temperature and humidity. A centralized dashboard with near real-time visualization and alerts makes it suitable for Smart Cities.
	Name of the Project under which technology is developed	Development of systems and solutions for smart cities using Internet of Things
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	The Air Quality Monitoring Station is designed to provide accurate and real-time assessment of air pollution levels in urban and industrial areas. The system employs environmental sensors connected to a microcontroller. Data collected from sensors is processed and transmitted to a cloud server or local interface using wireless communication technologies like Wi-Fi/ cellular network. The system enables continuous monitoring of harmful air pollutants and presents the data in an easy-to-understand format using

Sr. No.	Parameters	Brief details
		graphs and alerts. This technology supports environmental awareness, helps authorities in pollution control planning, and assists researchers in analysing long-term air quality trends. The solution is scalable, cost-effective, and suitable for deployment in smart city and environmental surveillance applications.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Puneeth G.P., C-DAC Bangalore puneeth@cdac.in +91 9738251235
12.	**Brief details of Technology Developed**	INDUS-IoT (Innovation, Development and UpSkilling -Internet of Things) Kit is C-DAC indigenously designed and developed low-cost loT development platform for users to quickly build loT applications and prototype their lot based innovative ideas.
	Name of the Project under which technology is developed	Development of systems and solutions for smart cities using Internet of Things
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	The INDUS-IoT development board comes with on-board sensors (Temperature, humidity, light intensity, accelerometer, PDM Mic etc.), actuators, an advanced ARM microcontroller supporting DSP, onboard Wi-Fi, Bluetooth/BLE wireless communication, USB, CAN interfaces and on-board debugger. This kit, combined with its software ecosystem is an all-in-one solution for researchers and students for up-skilling themselves in the advanced and emerging technologies of Internet of things (IoT).
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Shrikrishna S Chippalkatti, C-DAC, Bangalore krishnac@cdac.in +91 9916461784
13.	**Brief details of Technology Developed**	Indigenous smart thermal camera with AI capabilities
	Name of the Project under which technology is developed	Development of smart vision sensors for road traffic and industrial applications under InTranSe Phase-II (ITS R&D programme)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	Indigenously developed General-Purpose Thermal Camera integrates cutting-edge AI/ML-capable hardware, enabling seamless deployment of custom applications across domains such as industrial monitoring, security, medical diagnostics, environmental sensing, and traffic management. It uses an uncooled micro-bolometer with interchangeable 6–19 mm lenses, operates at 30 FPS, and captures heat-based imagery to deliver robust performance beyond visible-light systems. The generic heterogeneous hardware platform includes a quad application processor, GPU, video codec, and integrated H.264/H.265 encoding, enabling scalable, real-time thermal data processing. A dedicated deep-learning processing unit

Sr. No.	Parameters	Brief details
		optimized for CNNs supports common AI frameworks and pre-trained models, enabling advanced analytics, predictive insights, and autonomous decision-making. Demonstrated traffic applications include vehicle presence detection for adaptive signal optimization and pedestrian detection.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Sathyanarayan Satheesh G, C-DAC Thiruvananthapuram 9496750350, satheeshg@cdac.in
14.	**Brief details of Technology Developed**	Indigenous smart thermal camera with AI capabilities
	Name of the Project under which technology is developed	Development of smart vision sensors for road traffic and industrial applications under InTranSe Phase-II (ITS R&D programme)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	Indigenously developed General-Purpose Thermal Camera integrates cutting-edge AI/ML-capable hardware, enabling seamless deployment of custom applications across domains such as industrial monitoring, security, medical diagnostics, environmental sensing, and traffic management. It uses an uncooled micro-bolometer with interchangeable 6–19 mm lenses, operates at 30 FPS, and captures heat-based imagery to deliver robust performance beyond visible-light systems. The generic heterogeneous hardware platform includes a quad application processor, GPU, video codec, and integrated H.264/H.265 encoding, enabling scalable, real-time thermal data processing. A dedicated deep-learning processing unit optimized for CNNs supports common AI frameworks and pre-trained models, enabling advanced analytics, predictive insights, and autonomous decision-making. Demonstrated traffic applications include vehicle presence detection for adaptive signal optimization and pedestrian detection.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Sathyanarayan Satheesh G, C-DAC Thiruvananthapuram 9496750350, satheeshg@cdac.in
15.	**Brief details of Technology Developed**	Crowdsource Adaptive traffic signal control and optimization system
	Name of the Project under which technology is developed	Transfer of Technology Support (ToT Support)
	Funding Agency	Internal
	Technology Readiness Level (TRL)	7
	Brief Write-up on the Technology Developed	Computes signal timing generated from crowdsource data. Proposed signal timing by analysing travel time congestion level from crowdsource data. The outcome of the system indicates smoother traffic flow and improved signal efficiency, resulting in shorter waiting times for road users,

Sr. No.	Parameters	Brief details
		improving junction throughput. It improves Corridor-level performance with reduced delays and shorter queues along the entire corridor, leading to more consistent travel times.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Benoy Gopal E B , 9447057046, benoy@cdac.in Satheesh G, 9496750350, satheeshg@cdac.in
16.	**Brief details of Technology Developed**	Vissim based Simulation Interface for Adaptive Traffic Signal Control System
	Name of the Project under which technology is developed	Transfer of Technology Support (ToT Support)
	Funding Agency	Internal
	Technology Readiness Level (TRL)	7
	Brief Write-up on the Technology Developed	SimBridge is a software tool for evaluating C-DAC Adaptive Traffic Control System CoSiCoSt using traffic simulator VISSIM. This software tool is used to evaluate the signal plan generated for junctions in a corridor using micro-simulator VISSIM. It helps traffic engineers to simulate the junction plans for a corridor before deploying into the real field. SimBridge helps to identify the problem and select the right course of action for authorities, engineering consultants and researchers.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Benoy Gopal E B , 9447057046, benoy@cdac.in Satheesh G, 9496750350, satheeshg@cdac.in C-DAC Thiruvananthapuram
17.	**Brief details of Technology Developed**	CG-STAT, Potentiostat system based on Raspberry Pi and DAQ module for quality estimation of fermented food.
	Name of the Project under which technology is developed	e-GUNA: Sensory Assessment for Quality Estimation of Fermented Food
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-7
	Brief Write-up on the Technology Developed	A high-throughput field portable miniaturized Potentiostat with stabilized enzymatic bioreceptor toward cost effective cyanogenic glycoside detection in fermented food: special reference to bamboo shoot. Cyanogenic glycosides, present in bamboo shoot may lead to acute intoxications in humans, characterized: Growth retardation and neurological symptoms resulting from tissue damage in the central nervous system (CNS). CG-STAT is equipped with stable biosensor based screen printed electrode for DPV analysis and cyanogenic glycoside quantification in real samples.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Subrata Sarkar, +91 9433111038, subrata.sarkar@cdac.in Dr Subhankar Mukherjee, 9477355692, subhankar.mukherjee@cdac.in C-DAC Kolkata
18.	**Brief details of Technology Developed**	Computer-assisted autism screening and assessment system for early identification of Autism Spectrum Disorder (ASD) using standardized behavioral tools and digital analytics.

Sr. No.	Parameters	Brief details
	Name of the Project under which technology is developed	DISAAA – Development of an Integrated System for Autism Assessment and Analysis.
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-7
	Brief Write-up on the Technology Developed	DISAAA is a digital autism assessment platform that integrates standardized diagnostic tools, behavioral questionnaires and multimedia-based interaction modules to assist clinicians and psychologists in early screening and assessment of Autism Spectrum Disorder. The system supports structured data capture, automated scoring, analytics and report generation, enabling objective, consistent and scalable autism assessment in clinical and community settings.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr Kunal Chanda, 9830754257, kunal.chanda@cdac.in Mr Washef Ahmed, 6290023747, washef.ahmed@cdac.in C-DAC Kolkata
19.	**Brief details of Technology Developed**	A Wireless Early Fire Detection System AGNIVESH is a standalone system with thefacilities of data storage, data visualization andalarm notification.System comprises of battery poweredsensor modules, wireless routers and agateway for data aggregation.
	Name of the Project under which technology is developed	NA
	Funding Agency	NA
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	**Key Features** • Near real-time fire detection • Portable low power nodes • Long battery backup • Zigbee communication • Event-based alerts • Dashboard visualization
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dhivya G, C-DAC Chennai dhivyag@cdac.in 9789944905
20.	**Brief details of Technology Developed**	Development of a Meter Data Management System (MDMS) for secure storage, validation, processing, and analysis of smart meter data, supporting integration with Head End Systems and downstream applications.
	Name of the Project under which technology is developed	Smart City - Development of Meter Data Management System (MDMS) for Smart Energy Metering under Advanced Metering Infrastructure
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6

Sr. No.	Parameters	Brief details
	Brief Write-up on the Technology Developed	The Meter Data Management System (MDMS) is a critical component of the Advanced Metering Infrastructure (AMI) ecosystem, designed to collect, validate, store, and manage large volumes of smart meter data. The system performs data validation, estimation, and editing (VEE), supports interval and billing data processing, and ensures secure data handling.
	Details of Contact Person alongwith Mobile No. and e-mail ID	**Name:** Dr Karthika V, C-DAC Bangalore **Mobile No.:** 9844334898 **E-mail ID:** vkarthika@cdac.in
21.	**Brief details of Technology Developed**	Design and development of a scalable **Head End System (HES)** for smart energy meters, enabling secure device communication, remote meter management, data acquisition, event monitoring, and integration with Meter Data Management System (MDMS).
	Name of the Project under which technology is developed	Smart City - Advanced Metering Infrastructure (AMI) Implementation for Smart Energy Metering
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6 – System prototype deployed on cloud infrastructure and successfully demonstrated with smart meters in a campus environment.
	Brief Write-up on the Technology Developed	The Head End System (HES) is a core component of the Advanced Metering Infrastructure (AMI) ecosystem, designed to manage large-scale deployments of smart energy meters. The system supports secure two-way communication, enabling real-time and scheduled data collection, remote firmware upgrades, meter configuration, and event/alarm management. The HES ensures high reliability through robust retry mechanisms, fault detection, and automated data retransmission in case of communication failures. It adheres to relevant industry standards and interoperability guidelines, ensuring seamless integration with downstream systems such as MDMS, billing, and analytics platforms.
	Details of Contact Person alongwith Mobile No. and e-mail ID	**Name:** Dr Karthika V, C-DAC Bangalore **Mobile No.:** 9844334898 **E-mail ID:** vkarthika@cdac.in
22.	**Brief details of Technology Developed**	The Smart Water Meter is based on the Ultrasonic flow measurement and in-built LoRa wireless communication and it is primarily intended for domestic water distribution. A server/cloud based dashboard with complete water distribution system visualization makes it suitable for Smart Cities.
	Name of the Project under which technology is developed	Development of systems and solutions for smart cities using Internet of Things
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	A Smart Ultrasonic Water Meter with inbuilt LoRa communication provides accurate, low-maintenance water measurement using ultrasonic sensing with no moving parts. An embedded controller processes flow data and transmits it over long distances using low-power LoRa/LoRaWAN,

Sr. No.	Parameters	Brief details
		enabling multi-year battery operation. Features such as near real-time measurement and remote monitoring make it well suited for smart utility and smart city water management systems. A web-based interface and smart dashboard provide city administrators and citizens with actionable insights, enabling efficient water management, and data-driven decision-making for smarter urban governance.
	Details of Contact Person along with Mobile No. and e-mail ID	Mr. Puneeth G.P., C-DAC Bangalore puneeth@cdac.in +91 9738251235
23.	**Brief details of Technology Developed**	Smart Post Kiosk for any-time and any-where availability of postal services
	Name of the Project under which technology is developed	Development of systems and solutions for smart cities using Internet of Things
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	The Smart Post Kiosk (SPK) enables India Post to offer Speed Post and Registered Post services outside post offices and beyond working hours through a fully automated, self-service system available 24×7. Installed at public locations such as airports, stations, malls, and campuses, the kiosk allows users to enter sender and receiver details via a touch interface, generate a barcode, scan and deposit the article, and automatically calculate postage based on weight and service type. Payments are made digitally via any UPI app., after which the article is securely stored and an acknowledgement receipt is issued, along with email and SMS notifications. Mobile and web applications support pre-entry of details for faster use, while a centralized server-side dashboard enables administrators to remotely monitor, manage, and analyze kiosk operations and performance.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Pippala Kamala Krishna, C-DAC, Bangalore kamalakrishna@cdac.in +91 9945980670
24.	**Brief details of Technology Developed**	The technology is an integrated **IoT-based smart hydroponics system** designed for research, demonstration, and scalable deployment in protected cultivation environments such as polyhouses and greenhouses. The system enables **real-time monitoring, automation, and intelligent control** of critical hydroponic parameters including nutrient dosing, EC, pH, temperature, humidity, and circulation. It supports both **manual and automatic operating modes** and is suitable for leafy vegetables and high-value crops.
	Name of the Project under which technology is developed	Design and Deployment of IoT-based Hydroponics System for Agri-Research in the North East Region
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	6-7

Sr. No.	Parameters	Brief details
	Brief Write-up on the Technology Developed	The developed hydroponics system integrates **embedded systems, IoT, sensor networks, and automation software** into a single unified platform. The system architecture is centered around a **Raspberry Pi–based controller** with a user-friendly **GUI displayed on a touch-screen LCD**. It incorporates multiple actuators and sensors for precise control of the hydroponic environment. Key functional capabilities include: • Automated control of **nutrient dosing pumps** (multiple nutrients) based on EC and pH feedback • Control of **nutrient circulation pumps** with programmable schedules • Environmental control using **fan–pad system** for temperature regulation and **fogger** for humidity management • Real-time monitoring of **EC, pH, solution temperature, ambient temperature, and relative humidity** • Dual operating modes: ○ **Manual Mode:** User-defined set points for EC, pH, temperature, and humidity ○ **Automatic Mode:** Predefined crop-specific recipes and control logic • **Cloud connectivity** for remote monitoring, data logging, and visualization • Graphical visualization of historical and real-time data • Support for **online and offline operation** to ensure reliability in low-connectivity regions
	Details of Contact Person alongwith Mobile No. and e-mail ID	**Dr. Mandeep Singh,** C-DAC Mohali mandeep@cdac.in, +91 9463016065
25.	**Brief details of Technology Developed**	oneM2M based Common Service Layer Release 2
	Name of the Project under which technology is developed	Development of oneM2M based common service layer under InTranSe Phase-II (ITS R&D programme)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	6
	Brief Write-up on the Technology Developed	Common Smart IoT Connectivity (CoSMiC) is a standards-compliant middleware platform deployed between IoT field devices and application servers to enable a secure, interoperable, and multi-vendor IoT/M2M ecosystem, particularly for Intelligent Transportation Systems and smart city applications. Built on global oneM2M standards, CoSMiC provides a common service layer with open interfaces, robust device and user management, and support for data sharing across multiple verticals, while exposing open datasets and APIs that encourage innovation and scalable solution development. The underlying oneM2M standard has been adopted as the National Standard for the IoT/M2M ecosystem in India and is being implemented across numerous Smart Cities.

Sr. No.	Parameters	Brief details
	Details of Contact Person alongwith Mobile No. and e-mail ID	Hemant Jeevan Magdum, C-DAC Thiruvananthapuram hemant@cdac.in Satheesh G, C-DAC Thiruvananthapuram 9496750350, satheeshg@cdac.in
26.	**Brief details of Technology Developed**	Personalized Transit Route Guidance System (PTRGS)
	Name of the Project under which technology is developed	Intelligent Transportation Systems Endeavor for Indian Cities (InTranSE Ph-II) - ITS R&D programme
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	A mobile application that provides passengers with optimal or customized routes based on real-time information about location of buses and passenger demand. Using this app, passengers can find bus stops, bus arrival time, and bus route information. The routes can be selected based on minimum travel time, cost, crowding, or maximum reliability.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Abey SA, 9995029936, abeysa@cdac.in Satheesh G, 9496750350, satheeshg@cdac.in C-DAC Thiruvananthapuram
27.	**Brief details of Technology Developed**	Operational Strategies for Headway Reliability (OSHR)
	Name of the Project under which technology is developed	Intelligent Transportation Systems Endeavor for Indian Cities (InTranSE Ph-II) - ITS R&D programme
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	A real-time scheduling tool to improve transit reliability by reducing the number of bus bunching instances. This tool is an Android mobile application which uses GPS data of all the buses on a route and historic data to determine how long buses should be made to wait at control points (which are typically the start and end points on a route). These calculations are based on the predicted travel times between different stops on a route for different times of the day
	Details of Contact Person alongwith Mobile No. and e-mail ID	Abey SA, 9995029936, abeysa@cdac.in Satheesh G, 9496750350, satheeshg@cdac.in C-DAC Thiruvananthapuram
28.	**Brief details of Technology Developed**	Desktop Driving Simulator for Non-Lane based Mixed Traffic System
	Name of the Project under which technology is developed	Intelligent Transportation Systems Endeavor for Indian Cities (InTranSE Ph-II) - ITS R&D programme

Sr. No.	Parameters	Brief details
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	A user-friendly simulator's main goal is to provide a low-cost, easy-to-use driving simulator that will help with driving practice, driver tests, and licensing purposes. This simulator is exclusively designed for Indian road conditions and it is capable of simulating a typical Indian non lane based mixed traffic system. This is an interactive driving simulator integrated with a general-purpose traffic simulator which equips the user to drive in different visibility situations, practise parking practice sessions and enhance overall driving skills by providing detailed performance reports.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Abey SA, 9995029936, abeysa@cdac.in Satheesh G, 9496750350, satheeshg@cdac.in C-DAC Thiruvananthapuram
29.	**Brief details of Technology Developed**	**UltraSmart: The Multipurpose Smart Ultrasonic Device** Ultrasmart is a state-of-the-art, sensor-integrated IoT device developed by the Centre for Development of Advanced Computing (C-DAC), Hyderabad. Designed with flexibility, reliability, and low-power operation at its core, Ultrasmart addresses the growing demand for intelligent sensing in smart city and industrial environments. Its primary function is based on ultrasonic distance measurement, making it suitable for various applications such as parking systems, waste management, water level monitoring, and more. In addition to ultrasonic sensing, Ultrasmart integrates a temperature sensor and a 3-axis accelerometer, enhancing its ability to capture environmental and positional data. The device leverages LoRaWAN technology for long-range, low-power wireless communication, allowing seamless deployment across geographically dispersed sites.
30.	**Name of the Project under which technology is developed**	Smart City 2.0 Empowering Indian Industry with Smart Solutions
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	**Device Overview** Ultrasmart is a compact, multi-sensor IoT device built around an ultrasonic measurement module, with auxiliary environmental and motion sensors. The device is designed for low maintenance and long-duration field operation, with key features including: **1. Ultrasonic Distance Measurement** At the heart of Ultrasmart is an ultrasonic sensor capable of accurately detecting distances to nearby objects or surfaces. This is critical for monitoring the fill level in containers (e.g., garbage bins or water tanks), detecting obstructions, or measuring clearances in parking and automation systems. **2. Environmental and Motion Sensors**

Sr. No.	Parameters	Brief details
		• **Temperature Sensor:** Measures ambient temperature to support applications where environmental conditions affect operational thresholds (e.g., water systems or smart agriculture). • **Accelerometer:** A built-in 3-axis accelerometer detects movement, tilt, or impact, enabling tamper alerts, orientation detection, and motion-triggered actions. **3. LoRaWAN Communication** The device uses LoRaWAN (Long Range Wide Area Network) protocol to transmit sensor data to remote gateways or cloud platforms. This enables reliable, low-power communication over distances up to several kilometers, with minimal bandwidth usage i.e. ideal for large-scale IoT deployments. **Use Cases** 1. **Smart Parking** Detects vehicle presence to monitor bay occupancy and guide drivers in real time. 2. **Waste Bin Monitoring** Tracks fill levels and alerts for timely waste collection, reducing overflow and route inefficiency. 3. **Water Level Monitoring** Measures tank or reservoir levels for automated pump control and leak detection. **Salient Features** • **Wide Detection Range:** Covers 2-800 cm with minimal blind spots, ensuring accurate proximity detection. • **Easy Installation:** Adaptable to various applications, making it highly versatile. • **Temperature Sensor:** Detects temperature fluctuations • **3-Axis Accelerometer**: Monitors tilt to keep track of the device's orientation • **GNSS**: For Tracking (***Optional)*** • **High Durability:** IP67-rated waterproof enclosure designed to withstand outdoor conditions. • **LoRaWAN Compatibility:** Works seamlessly with standard LoRaWAN gateways and network servers for reliable, long-range connectivity. • **Extended Battery Life*:** up to 10 years of operation ***Based on application requirements***
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Sunnam Venkata Srikanth, C-DAC Hyderabad 9949693448 venkatas@cdac.in
31.	**Brief details of Technology Developed**	BioSTAT, a development of a three-electrode potentiostat system capable of measuring ultra-low currents down to 32 nA. The system supports both screen-printed electrodes (SPEs) and conventional probe-based electrodes through appropriate connector interfaces.

Sr. No.	Parameters	Brief details
	Name of the Project under which technology is developed	SWISS: Safe Water Index Sense using biochemical Sensor
	Funding Agency	Department of Biotechnology, Ministry of Science & Technology
	Technology Readiness Level (TRL)	TRL-6
	Brief Write-up on the Technology Developed	The device is capable of detecting heavy metals in water samples. As developed system is capable of measuring ultra-low currents, up to 32 nA, generated during voltammetric analysis, the device is suitable for a wide range of applications, including pesticide detection in fruits and vegetables, investigation of enzymatic reactions that result in changes in the concentration of charged species in solution, and analysis of pathological samples where precise measurement of ionic variations is critical
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Subrata Sarkar, +91 9433111038, subrata.sarkar@cdac.in Dr Subhankar Mukherjee, 9477355692, subhankar.mukherjee@cdac.in C-DAC Kolkata
32.	**Brief details of Technology Developed**	Development of Algorithms and Electronics for Medical Grade Dosimeters. Development of Vega based interfacing with Analog Front End. Development of Low Current Measurement Circuits. Application developed for Dosimeter Networking, Data visualization and Networking. Hardware development of the Network Hub
	Name of the Project under which technology is developed.	Development of Indigenous Medical Radiation Dosimeter based on Organic field effect transistors (DosFET)
	Funding Agency	Ministry of Electronics and Information Technology
	Technology Readiness Level (TRL)	5+
	Brief Write-up on the Technology Developed	We have developed an innovative and crucial methodology and device for the application of Medical grade dosimetry, The System , Sub-System and devices are designed to measure the intensity of radiation from radiotherapy and generate analytics for safe limit of exposure for the professionals involved in the operational activities. The Dosimeter is being developed based on Silicon PIN Photodiodes with radiant sensitivity which can respond to the radiation falling on to the diode which can generate electrical signals and can be processed with our developed electronics. We have also collaboratively worked for a development of an indigenous sensor and analog front end for precise radiation measurement. In this project we have also developed a Network Hub The network of devices are managed using a centralized network hub for analytics and alert generation.
	Details of Contact Person along with Mobile No. and e-mail ID	Shri. Jitesh Choudhary, C-DAC CINE +91 9441927003 jitesh@cdac.in

Sr. No.	Parameters	Brief details
33.	**Brief details of Technology Developed**	A complete in-vehicle Public Key Infrastructure security system has been developed to cryptographically authenticate critical automotive components and control units in the vehicle. The framework implements a hierarchical trust model with a Root Certificate Authority, an intermediate In-Vehicle Certificate Authority, and device-level certificates assigned to each component. Unique, unforgeable digital identities are managed by the Certifying Authority and validated in real-time before permitting critical command execution over the vehicle protocols. The proof of concept establishes a production-aligned security layer that prevents unauthorized access and command injection, creating a trusted foundation for secure in-vehicle communication and over-the-air updates.
	Name of the Project under which technology is developed	Development of PKI-based Digital Certificates for IoT Device Security with PoC for Automotive Industry
	Funding Agency	Controller of Certifying Authority, Ministry of Electronics and Information Technology,
	Technology Readiness Level (TRL)	4
	Brief Write-up on the Technology Developed	A comprehensive Public Key Infrastructure security framework has been developed for automotive in-vehicle networks, establishing cryptographic trust across vehicle electronic systems. This technology implements a hierarchical certificate authority model within the vehicle that functions as an intermediate Certificate Authority, managing the complete lifecycle of X.509 digital certificates for Automotive components. The system enables mutual authentication between vehicle components through certificate-based validation, with hardware secure elements providing tamper-resistant key storage and cryptographic acceleration for operations including elliptic curve digital signatures and certificate signing request generation. The system is hardened with hardware-based root of trust via integrated secure elements, including Infineon TPM and Optiga Trust M chips, which provide tamper-resistant key storage and cryptographic acceleration. Furthermore, we have developed a Secure OTA Update Pipeline that cryptographically signs firmware, distributes it via targeted Invehicle protocol based messaging, and validates installation across the vehicle network. This end-to-end framework—encompassing certificate lifecycle management, real-time GUI monitoring.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Karthika V., C-DAC Bangalore 9844334898 vkarthika@cdac.in
34.	**Brief details of Technology Developed**	Design, development, and validation of a Time-Sensitive Networking (TSN) based deterministic communication framework for latency-critical Smart City applications such as Intelligent Transportation Systems (ITS), real-time surveillance, smart grids, and mission-critical IoT services.
	Name of the Project under which technology is developed.	Smart City 2.0 (under AMRUT 2.0 initiative)
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL 3-4

Sr. No.	Parameters	Brief details
	Brief Write-up on the Technology Developed	The developed technology focuses on enabling deterministic, low-latency, and high-reliability communication using IEEE Time-Sensitive Networking (TSN) standards such as IEEE 802.1AS (gPTP), 802.1Qbv (Time-Aware Shaper), and 802.1Qci (Per-Stream Filtering and Policing). The solution ensures precise time synchronization (nanosecond-level accuracy), bounded latency, and traffic prioritization across heterogeneous network devices. The TSN framework is targeted for Smart City 2.0 use cases including traffic signal coordination, emergency response systems, industrial IoT integration.
	Details of Contact Person along with Mobile No. and e-mail ID	Mr. Jitesh Choudhary +91 9441927003 jitesh@cdac.in
35.	**Brief details of Technology Developed**	IoT device functionality sandbox provides a secure environment for testing and validating IoT devices before they are deployed in the field.
	Name of the Project under which technology is developed	A Comprehensive IoT Security Ecosystem and Sandbox
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	NA
	Brief Write-up on the Technology Developed	We are developing an end-to-end testing environment for IoT devices that includes important tests like functionality of the embedded protocol testing (UART, SPI, I2C and, CAN), SAR testing for human safety, EMC testing for electromagnetic interference, product safety validation, and protocol compatibility/interoperability testing across numerous communication protocols. By executing these tests in a controlled environment, any failures, non-compliance concerns, and interoperability challenges may be recognised at an early stage. The centralized web portal is a significant part of the developed technology. It lets you access testing procedures, real-time monitoring, data logging, and report generating all from one place.
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Mr. Raghu H V, C-DAC Bangalore raghuhv@cdac.in +91 9986215591